Power Transmission Systems & Line Performance

PCEOT503: Power Systems (S5 EEE)

Bachelor of Technology (B.Tech) — 2024 Scheme

Complete Technical Notes, Step-by-Step Derivations & Numerical Problem Suite

Prepared from Course Materials, Lecture Slides & KTU Curriculum

Module 2 Complete Syllabus Coverage:

- Electrical Model & Line Parameters: Resistance, Inductance & Capacitance
- Rigorous Derivation of Three-Phase Double Circuit Lines (GMD & GMR Methods)
- Transmission Line Modelling: Short, Medium (Nominal-T & π), and Long Lines
- Derivation and Calculation of Two-Port ABCD Parameters & Voltage Regulation
- Operational Phenomena: Skin Effect, Ferranti Effect, Corona & Surge Impedance Loading
- Overhead Insulators: String Efficiency, Potential Grading & 12 Numerical Problems

Compiled August 17, 2026

Contents

1 Transmission Line Parameters (Electrical Model) 2
1.1 Physical Nature of Transmission Line Parameters 2
1.2 Conductor Materials and Types 2
1.3 Conductor Resistance 3
1.4 Line Inductance Fundamentals 3
1.5 Inductance of 3-Phase Single-Circuit Lines 5
1.6 Derivation of Inductance for Three-Phase Double-Circuit Lines 6
1.7 Line Capacitance Fundamentals 7
1.8 Derivation of Capacitance for Three-Phase Double-Circuit Lines 9

2 Transmission Line Modelling and Two-Port ABCD Parameters 11
2.1 Classification of Transmission Lines 11
2.2 The Transmission Line as a Two-Port Network 11
2.3 Derivation of ABCD Parameters for Line Models 12
2.4 Voltage Regulation and Transmission Efficiency 14

3 Operational Phenomena in Transmission Lines 15
3.1 The Skin Effect 15
3.2 The Ferranti Effect 15
3.3 Corona Phenomenon (Qualitative Study) 16
3.4 Surge Impedance Loading (SIL) 17

4 Overhead Line Insulators and String Efficiency 18
4.1 Insulator Types and Mechanical Requirements 18
4.2 Potential Distribution Across a Suspension Insulator String 18
4.3 String Efficiency and Methods of Improvement 19

5 Comprehensive Numerical Problem Suite 20

6 Quick Revision Sheet & Model Examination Questions 27
6.1 Master Formula Summary Table for Module 2 27
6.2 Model Examination Questions (KTU S5 Pattern) 27

1 Transmission Line Parameters (Electrical Model)

Syllabus Requirement: Power Transmission System - (Electrical Model) - Line parameters – resistance - inductance and capacitance (Derivation of three phase double circuit)

1.1 Physical Nature of Transmission Line Parameters

An overhead power transmission line is not merely a pair of ideal zero-impedance conductors. Instead, it is an electromechanical distributed network characterized by four fundamental distributed primary constants:

1. **Series Resistance (R):** Arises from the ohmic resistivity of the conductor material, dissipating energy as heat (I^2R losses).
2. **Series Inductance (L):** Arises from magnetic flux linkages set up inside and around current-carrying conductors.
3. **Shunt Capacitance (C):** Arises from the electric potential difference between energized conductors and between conductors and the ground plane.
4. **Shunt Conductance (G):** Arises from leakage currents over insulator surfaces and atmospheric corona discharge (usually neglected under normal clean weather conditions).

These distributed line parameters determine the extbfvoltage drop, voltage regulation, power transfer capability, efficiency, and transient stability of the power grid.

1.2 Conductor Materials and Types

1. Solid vs. Stranded Conductors

Solid conductors are rigid and prone to severe mechanical fatigue cracking from wind-induced aeolian vibrations. Modern transmission systems deploy extbfstranded conductors, consisting of multiple thin wires helically twisted together to offer superior mechanical flexibility, high tensile strength, and easier transportation on cable drums.

2. ACSR (Aluminum Conductor Steel Reinforced)

ACSR is the global standard conductor for overhead high-voltage transmission.

- **Central Core:** High-strength galvanized steel strands provide high tensile strength and minimize line sag between towers.
- **Outer Layers:** High-purity electrical grade aluminum (EC grade, $> 61\%$ IACS conductivity) carry the electrical current.
- **Notation:** ACSR is designated by strand count, e.g., $54/7$ represents 54 aluminum strands surrounding 7 steel core strands.

3. Bundled Conductors

For Extra High Voltage (EHV ≥ 400 kV) and Ultra High Voltage (UHV) systems, each phase conductor is split into a bundle of two, three, four, or more sub-conductors separated by mechanical spacers ($d = 30 - 50$ cm).

- **Advantages of Bundling:**
  1. Drastically increases the effective **Self Geometric Mean Radius (GMR)**, which reduces the line series inductive reactance: $X_L \propto \ln(D_m/D_s)$.
  2. Significantly increases line shunt capacitance, which boosts the **Surge Impedance Loading (SIL)**: SIL $= V^2/Z_c$.
  3. Reduces the surface electrostatic voltage gradient (E_{surface}), thereby drastically reducing or eliminating **Corona loss, audible hissing noise, and Radio/TV Interference (RI/TVI)**.

1.3 Conductor Resistance

DC Resistance and Temperature Dependency

The direct-current (DC) resistance of a solid cylindrical conductor of length l and cross-sectional area A is given by:

$$R_{\text{dc}} = \frac{\rho \cdot l}{A} \quad [\Omega] \tag{1}$$

where ρ is the resistivity of the material ($\rho_{\text{copper}} \approx 1.72 \times 10^{-8}\,\Omega \cdot \text{m}$, $\rho_{\text{aluminum}} \approx 2.83 \times 10^{-8}\,\Omega \cdot \text{m}$ at $20°\text{C}$).

Resistance varies linearly with operating temperature according to:

$$R_{T_2} = R_{T_1}\left[1 + \alpha_{T_1}(T_2 - T_1)\right] = R_{T_1}\left(\frac{T_2 + T_0}{T_1 + T_0}\right) \tag{2}$$

where $\alpha_{20} \approx 0.00403\ /°\text{C}$ for aluminum, and T_0 is the inferred zero-resistance temperature ($-228°\text{C}$ for aluminum, $-234.5°\text{C}$ for annealed copper). For ampacity design, resistance is evaluated at maximum operating temperature ($75°\text{C}$ or $85°\text{C}$).

Effective AC Resistance and Stranding Factor

Under alternating current (50 Hz), two additional physical factors increase the effective resistance:

1. **Skin Effect:** Non-uniform current distribution forces AC current toward the outer periphery, reducing effective cross-sectional area and causing $R_{\text{ac}} > R_{\text{dc}}$ (typically $1.02 - 1.08 \times R_{\text{dc}}$ at 50 Hz).
2. **Stranding / Spiraling Lay Effect:** Helically spiraled outer strands are slightly longer than the linear axial length of the cable, adding approximately **2%** to the effective resistance.

For a double-circuit line where each phase consists of two identical circuits in parallel, the total equivalent resistance per phase is:

$$R_{\text{phase}} = \frac{R_{\text{circuit}}}{2} \tag{3}$$

1.4 Line Inductance Fundamentals

Magnetic Field & Flux Linkages

The inductance of an electrical circuit is defined as the total magnetic flux linkages per ampere of current:

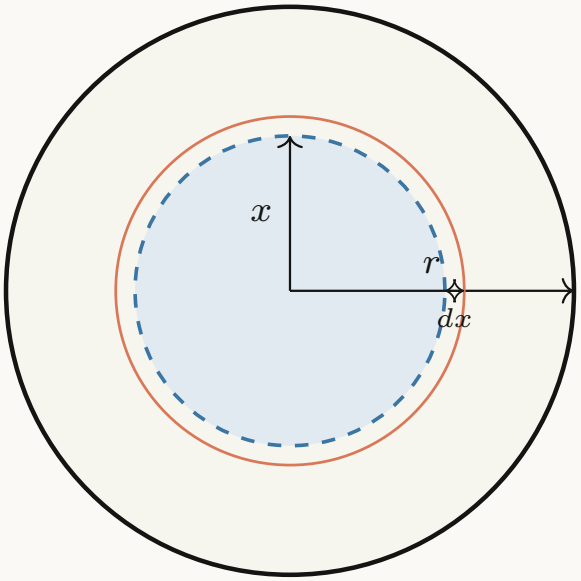
$$L = \frac{\lambda}{I} = \frac{N\Phi}{I} \quad [\text{Henries/meter}] \tag{4}$$

For an isolated cylindrical conductor of radius r carrying a steady uniform current I, the magnetic flux linkages consist of two distinct spatial components:

$$\lambda_{\text{total}} = \lambda_{\text{internal}} + \lambda_{\text{external}} \tag{5}$$

Derivation of Internal Flux Linkage (λ_{int})

Consider a long cylindrical conductor of radius r carrying current I uniformly distributed with current density $J = \frac{I}{\pi r^2}$.



Internal Shell of Radius x and Thickness dx

1. By Ampere's Circuital Law, the magnetic field intensity H_x at an internal radius $x \leq r$ enclosed by the path is:

$$\oint H_x \cdot dl = I_x \implies H_x(2\pi x) = I\left(\frac{\pi x^2}{\pi r^2}\right) = I\left(\frac{x^2}{r^2}\right) \implies H_x = \frac{Ix}{2\pi r^2} \quad \text{[A/m]} \tag{6}$$

2. The magnetic flux density is $B_x = \mu_0 H_x = \frac{\mu_0 I x}{2\pi r^2}$ (assuming non-magnetic conductor $\mu_r = 1, \mu_0 = 4\pi \times 10^{-7}$ H/m).
3. The differential flux per meter length in the tubular shell of thickness dx is $d\Phi = B_x \cdot (1 \cdot dx) = \frac{\mu_0 I x}{2\pi r^2} dx$.
4. This flux links only the fraction of current inside radius x, i.e., $N_x = \frac{\pi x^2}{\pi r^2} = \frac{x^2}{r^2}$. The internal flux linkage is:

$$\lambda_{\text{int}} = \int_0^r \left(\frac{x^2}{r^2}\right) d\Phi = \int_0^r \frac{\mu_0 I x^3}{2\pi r^4} dx = \frac{\mu_0 I}{2\pi r^4}\left[\frac{x^4}{4}\right]_0^r = \frac{\mu_0 I}{8\pi} \quad \text{[Wb-t/m]} \tag{7}$$

5. Hence, the internal inductance is constant and extbfindependent of conductor radius:

$$L_{\text{int}} = \frac{\lambda_{\text{int}}}{I} = \frac{\mu_0}{8\pi} = \frac{4\pi \times 10^{-7}}{8\pi} = \frac{1}{2} \times 10^{-7} \quad \text{[H/m]} \tag{8}$$

Derivation of External Flux Linkage (λ_{ext})

For the magnetic flux outside the conductor between the surface r and an external distance D:

$$H_x(2\pi x) = I \implies B_x = \frac{\mu_0 I}{2\pi x} \tag{9}$$

Since all external flux links the entire current ($N = 1$), the external flux linkage is:

$$\lambda_{\text{ext}} = \int_r^D B_x dx = \int_r^D \frac{\mu_0 I}{2\pi x} dx = \frac{\mu_0 I}{2\pi} \ln\left(\frac{D}{r}\right) \quad \text{[Wb-t/m]} \tag{10}$$

The external inductance is:

$$L_{\text{ext}} = \frac{\mu_0}{2\pi} \ln\left(\frac{D}{r}\right) = 2 \times 10^{-7} \ln\left(\frac{D}{r}\right) \quad \text{[H/m]} \tag{11}$$

Total Inductance and the Fictitious Self-GMR Concept

Combining the internal and external components:

$$L_{\text{total}} = L_{\text{int}} + L_{\text{ext}} = 2 \times 10^{-7} \left[\frac{1}{4} + \ln\left(\frac{D}{r}\right)\right] \quad \text{[H/m]} \tag{12}$$

Using the mathematical identity $\frac{1}{4} = \ln(e^{1/4})$:

$$L_{\text{total}} = 2 \times 10^{-7} \left[\ln(e^{1/4}) + \ln\left(\frac{D}{r}\right)\right] = 2 \times 10^{-7} \ln\left(\frac{D}{r \cdot e^{-1/4}}\right) \tag{13}$$

Self GMR (r') of a Solid Conductor: We define the extbfeffective fictitious radius or extbf-Self Geometric Mean Radius (GMR / D_s) as:

$$r' = D_s = r \cdot e^{-1/4} = r \cdot e^{-0.25} \approx 0.7788 \cdot r \tag{14}$$

By replacing the physical solid conductor of radius r with an equivalent fictitious hollow, infinitely thin shell of radius r', the internal inductance is seamlessly accounted for in the single logarithmic expression:

$$L = 2 \times 10^{-7} \ln\left(\frac{D}{r'}\right) = 2 \times 10^{-7} \ln\left(\frac{D_m}{D_s}\right) \quad \text{[H/m]} \tag{15}$$

1.5 Inductance of 3-Phase Single-Circuit Lines

1. **Symmetrical / Equilateral Spacing ($D_{12} = D_{23} = D_{31} = D$):** Due to symmetry, the flux linkages of all three phases are balanced:

$$L_{\text{phase}} = 2 \times 10^{-7} \ln\left(\frac{D}{r'}\right) \quad \text{[H/m/phase]} \tag{16}$$

2. **Unsymmetrical Spacing with Transposition ($D_{12} \neq D_{23} \neq D_{31}$):** If conductors are not transposed, the three phases experience unequal induced voltages and unbalanced impedance drops. When the line is extbfregularly transposed (cyclically rotating conductor positions in 3 equal sections along the route), the average phase inductance is:

$$L_{\text{phase}} = 2 \times 10^{-7} \ln\left(\frac{D_{\text{eq}}}{r'}\right) = 2 \times 10^{-7} \ln\left(\frac{D_m}{D_s}\right) \quad \text{[H/m/phase]} \tag{17}$$

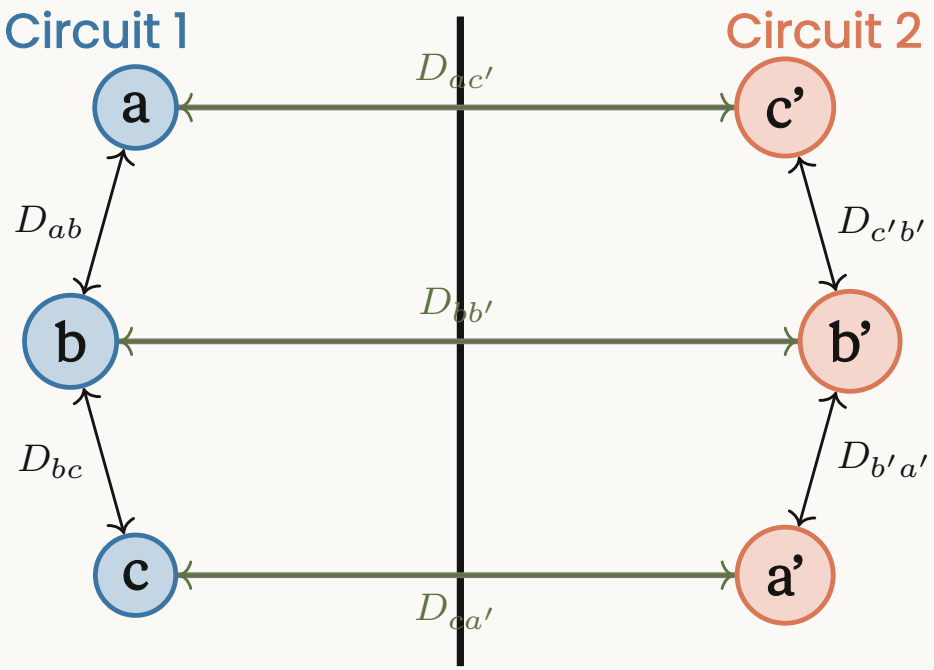
where the equivalent mutual geometric mean distance is:

$$D_{\text{eq}} = D_m = \sqrt[3]{D_{12} \cdot D_{23} \cdot D_{31}} \tag{18}$$

1.6 Derivation of Inductance for Three-Phase Double-Circuit Lines

Syllabus Requirement: Derivation of three phase double circuit line parameters (inductance and capacitance)

A three-phase double-circuit line consists of two parallel circuits sharing the same transmission tower structure: Circuit 1 (a, b, c) and Circuit 2 (a', b', c'). To maintain balanced electrical characteristics and minimize total loop inductance, the conductors are transposed cyclically and arranged in an inverted hexagonal/vertical configuration.



Complete Step-by-Step Derivation of Inductance for 3-Φ Double Circuit

Let each conductor have physical radius r, fictitious self-radius $r' = 0.7788r$, and let the line carry balanced 3-phase currents:

$$I_a + I_b + I_c = 0 \tag{19}$$

Because the two circuits operate in parallel, the current divides equally between paired conductors: $I_a = I_{a'} = \frac{I_A}{2}, I_b = I_{b'} = \frac{I_B}{2}, I_c = I_{c'} = \frac{I_C}{2}$.

Step 1: Evaluation of Self GMD (D_s / GMR of each phase)

For Phase A, which consists of parallel conductors a and a':

$$D_{sa} = \sqrt[4]{D_{aa} \cdot D_{aa'} \cdot D_{a'a} \cdot D_{a'a'}} = \sqrt[4]{r' \cdot D_{aa'} \cdot D_{a'a} \cdot r'} = \sqrt{r' \cdot D_{aa'}} \tag{20}$$

Similarly, for Phase B (conductors b, b') and Phase C (conductors c, c'):

$$D_{sb} = \sqrt{r' \cdot D_{bb'}} \tag{21}$$

$$D_{sc} = \sqrt{r' \cdot D_{cc'}} \tag{22}$$

The equivalent composite extbfSelf GMD (D_s) of the transposed three-phase system is the geometric mean:

$$D_s = \sqrt[3]{D_{sa} \cdot D_{sb} \cdot D_{sc}} = \sqrt[3]{\sqrt{r' D_{aa'}} \cdot \sqrt{r' D_{bb'}} \cdot \sqrt{r' D_{cc'}}} \tag{23}$$

Step 2: Evaluation of Mutual GMD (D_m between phase pairs)

The mutual distance between Phase A (a, a') and Phase B (b, b') is the geometric mean of all 4 mutual distances:

$$D_{ab} = \sqrt[4]{D_{ab} \cdot D_{ab'} \cdot D_{a'b} \cdot D_{a'b'}} \tag{24}$$

Similarly, between Phase B and Phase C, and between Phase C and Phase A:

$$D_{bc} = \sqrt[4]{D_{bc} \cdot D_{bc'} \cdot D_{b'c} \cdot D_{b'c'}} \tag{25}$$

$$D_{ca} = \sqrt[4]{D_{ca} \cdot D_{ca'} \cdot D_{c'a} \cdot D_{c'a'}} \tag{26}$$

The equivalent composite extbfMutual GMD (D_m) of the transposed three-phase system is:

$$D_m = \sqrt[3]{D_{ab} \cdot D_{bc} \cdot D_{ca}} \tag{27}$$

Step 3: Final Phase Inductance Formula

Applying the GMD-GMR method, the total inductance per phase per unit length is:

$$L_{\text{phase}} = 2 \times 10^{-7} \ln\left(\frac{D_m}{D_s}\right) \quad \text{[H/m/phase]} \tag{28}$$

Or equivalently in millihenries per kilometer:

$$L_{\text{phase}} = 0.2 \ln\left(\frac{D_m}{D_s}\right) \quad \text{[mH/km/phase]} \tag{29}$$

Why Inverted Phase Configuration ($a - b - c$ vs $c' - b' - a'$) is Preferred

Notice in the tower schematic that Circuit 1 has vertical arrangement $a - b - c$ while Circuit 2 is inverted to $c' - b' - a'$. This maximizes the diagonal separation between identical phases $(D_{aa'}, D_{cc'})$, which **maximizes the composite Self GMR (D_s)** and minimizes the mutual coupling, thereby reducing the total series inductive reactance by **10–20%** compared to non-inverted layouts.

1.7 Line Capacitance Fundamentals

Electric Field and Potential Difference

By Gauss's Law, a long cylindrical conductor of radius r carrying a uniform surface charge q C/m creates a radial electric field in surrounding dielectric medium of permittivity $\epsilon = \epsilon_0 \epsilon_r$ ($\epsilon_0 \approx 8.854 \times 10^{-12}$ F/m):

$$E_x = \frac{q}{2\pi\epsilon x} \quad \text{[V/m]} \tag{30}$$

The electric potential difference between radius r and an external point at distance D is:

$$V = \int_r^D E_x dx = \int_r^D \frac{q}{2\pi\epsilon x} dx = \frac{q}{2\pi\epsilon} \ln\left(\frac{D}{r}\right) \quad \text{[Volts]} \tag{31}$$

Capacitance of a Single-Phase 2-Wire Line

For two parallel conductors separated by center-to-center distance D carrying charges $+q$ and $-q$:

$$V_{ab} = \frac{q}{\pi\epsilon_0} \ln\left(\frac{D}{r}\right) \implies C_{ab} = \frac{q}{V_{ab}} = \frac{\pi\epsilon_0}{\ln(D/r)} \quad \text{[F/m]} \tag{32}$$

The line-to-neutral (or line-to-ground) capacitance is double the line-to-line value:

$$C_n = 2C_{ab} = \frac{2\pi\epsilon_0}{\ln(D/r)} \quad [\text{F/m}] \tag{33}$$

Critical Distinction: Radius in Inductance vs. Capacitance

In inductance calculations, the magnetic flux penetrating the conductor interior requires using the fictitious radius $r' = 0.7788r$. In capacitance calculations, electrostatic charge resides extbfstrictly on the outer physical conductor surface ($E = 0$ inside conductor), so extbfalways use the actual physical radius r (do NOT apply the 0.7788 factor!).

1.8 Derivation of Capacitance for Three-Phase Double-Circuit Lines

Complete Step-by-Step Derivation of Capacitance for 3-Φ Double Circuit

Consider a transposed 3-phase double-circuit line carrying charges q_a, q_b, q_c per phase such that:

$$q_a + q_b + q_c = 0 \tag{34}$$

The charge on each phase divides equally between the two parallel conductors: $q_a = q_{a'} = \frac{q_A}{2}, q_b = q_{b'} = \frac{q_B}{2}, q_c = q_{c'} = \frac{q_C}{2}$.

Step 1: Evaluation of Capacitive Self GMR (D_{sc})

For Phase A, using the actual conductor radius r:

$$D_{sca} = \sqrt{r \cdot D_{aa'}} \tag{35}$$

Similarly for Phases B and C:

$$D_{scb} = \sqrt{r \cdot D_{bb'}} \tag{36}$$

$$D_{scc} = \sqrt{r \cdot D_{cc'}} \tag{37}$$

The equivalent composite extbfCapacitive Self GMD (D_{sc}) of the transposed system is:

$$D_{sc} = \sqrt[3]{D_{sca} \cdot D_{scb} \cdot D_{scc}} = \sqrt[3]{\sqrt{rD_{aa'}} \cdot \sqrt{rD_{bb'}} \cdot \sqrt{rD_{cc'}}} \tag{38}$$

Step 2: Evaluation of Mutual GMD (D_m)

The mutual geometric mean distance D_m is identical to that used in inductance calculations:

$$D_{ab} = \sqrt[4]{D_{ab}D_{ab'}D_{a'b}D_{a'b'}}, \quad D_{bc} = \sqrt[4]{D_{bc}D_{bc'}D_{b'c}D_{b'c'}}, \quad D_{ca} = \sqrt[4]{D_{ca}D_{ca'}D_{c'a}D_{c'a'}} \tag{39}$$

$$D_m = \sqrt[3]{D_{ab} \cdot D_{bc} \cdot D_{ca}} \tag{40}$$

Step 3: Final Line-to-Neutral Capacitance Formula

The line-to-neutral capacitance per unit length of the transposed double-circuit line is:

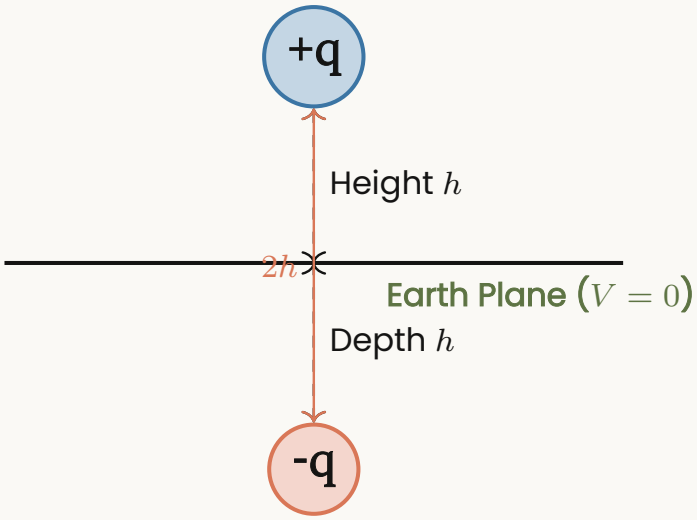
$$C_n = \frac{2\pi\epsilon_0}{\ln\left(\frac{D_m}{D_{sc}}\right)} \quad \text{[F/m/phase]} \tag{41}$$

Or equivalently in microfarads per kilometer:

$$C_n = \frac{0.0242}{\log_{10}\left(\frac{D_m}{D_{sc}}\right)} \quad [\mu\text{F/km/phase}] \tag{42}$$

Effect of Earth on Transmission Line Capacitance (Method of Images)

The earth surface acts as a conducting equipotential plane ($V = 0$). By Kelvin's extbfMethod of Images, the boundary condition at the earth plane is satisfied by replacing the earth with a fictitious image conductor at depth h carrying charge $-q$.



Including the image charges modifies the capacitance formula to:

$$C_n = \frac{2\pi\epsilon_0}{\ln\left(\frac{D}{r}\right) - \ln\left(\sqrt{1 + \left(\frac{D}{2h}\right)^2}\right)} \quad \text{[F/m]} \tag{43}$$

Because the denominator is slightly decreased, the presence of the earth extbfincreases the line-to-neutral capacitance (typically by $1-5\%$), though for typical high-voltage lines with large clearance ($h \gg D$), this effect is small.

2 Transmission Line Modelling and Two-Port ABCD Parameters

Syllabus Requirement: Transmission line modelling - classifications (concept only) – transmission line as two port network – derivation and calculation of ABCD parameters (derivation and numerical problems)

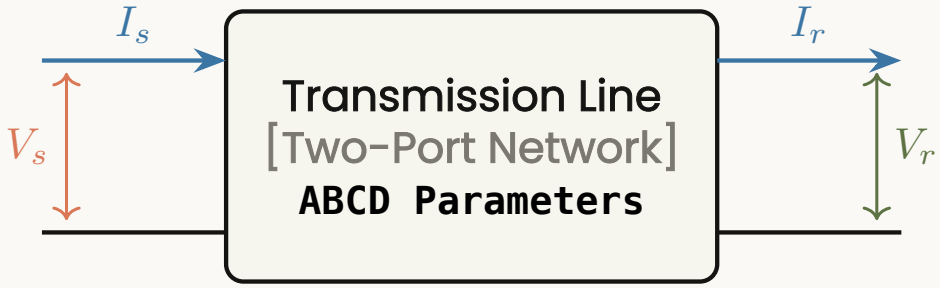
2.1 Classification of Transmission Lines

Transmission lines are classified into three distinct categories based on their physical length l and operating voltage V:

Classification	Length (l)	Voltage (V)	Capacitance Treatment	Mathematical Model
Short Line	< 80 km	< 20 kV	Neglected entirely	Simple lumped series $R-L$ impedance
Medium Line	$80 - 250$ km	$20 - 100$ kV	Lumped at discrete nodes	Nominal-T or Nominal-π networks
Long Line	> 250 km	> 100 kV	Uniformly distributed	Rigorous distributed hyperbolic equations

2.2 The Transmission Line as a Two-Port Network

A transmission line is a linear, passive, and bilateral electrical network with two input terminals (Sending End, subscript s) and two output terminals (Receiving End, subscript r).



The Governing Matrix Equations

$$\begin{bmatrix} V_s \\ I_s \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_r \\ I_r \end{bmatrix} \tag{44}$$

$$V_s = AV_r + BI_r \tag{45}$$

$$I_s = CV_r + DI_r \tag{46}$$

Physical Definitions & Dimensions of ABCD Parameters

1. $A = \left.\frac{V_s}{V_r}\right|_{I_r=0}$: extbfOpen-circuit reverse voltage ratio (Dimensionless).
2. $B = \left.\frac{V_s}{I_r}\right|_{V_r=0}$: extbfShort-circuit transfer impedance (Units: Ohms, Ω).
3. $C = \left.\frac{I_s}{V_r}\right|_{I_r=0}$: extbfOpen-circuit transfer admittance (Units: Siemens or $\mho$).

4. $D = \left.\frac{I_s}{I_r}\right|_{V_r=0}$: extbfShort-circuit reverse current ratio (Dimensionless).

Fundamental Network Conditions

- **Condition for Symmetry:** A transmission line looks identical viewed from either terminal pair:

$$A = D \tag{47}$$

- **Condition for Reciprocity:** For any linear, bilateral, passive network:

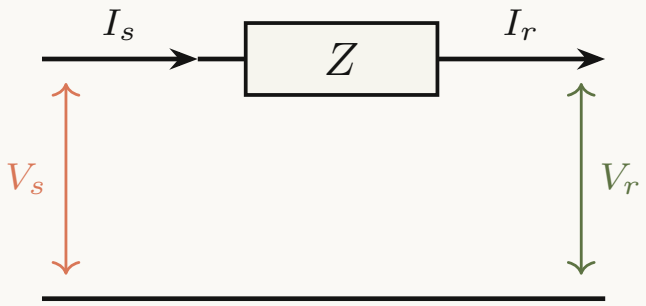
$$AD - BC = 1 \tag{48}$$

2.3 Derivation of ABCD Parameters for Line Models

Let total series line impedance be $Z = R + jX_L = R + j\omega L$, and total line shunt admittance be $Y = G + jB_C = j\omega C$ (neglecting conductance $G = 0$).

1. Short Transmission Line Model (< 80 km)

In short lines, shunt capacitance is negligible ($Y = 0$).



Applying KVL and KCL:

$$V_s = V_r + I_r Z = (1)V_r + (Z)I_r \tag{49}$$
$$I_s = I_r = (0)V_r + (1)I_r \tag{50}$$

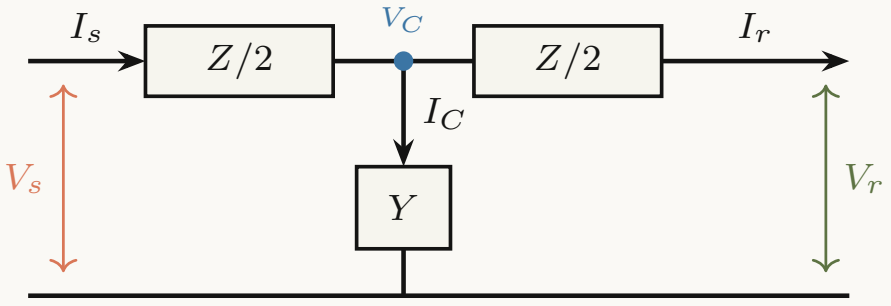
Comparing with standard ABCD equations:

$$A = 1, \quad B = Z, \quad C = 0, \quad D = 1 \tag{51}$$

Verification: $AD - BC = (1)(1) - (Z)(0) = 1$ ✓

2. Medium Line: Nominal-T Method ($80 - 250$ km)

In the Nominal-T network, total series impedance is split equally into two halves ($Z/2$), and the entire shunt admittance Y is lumped at the center node.



1. Voltage at central junction: $V_C = V_r + I_r\left(\frac{Z}{2}\right)$.
2. Current through central shunt capacitor: $I_C = YV_C = Y\left[V_r + I_r\left(\frac{Z}{2}\right)\right] = YV_r + \frac{YZ}{2}I_r$.
3. Sending-end current:

$$I_s = I_r + I_C = YV_r + \left(1 + \frac{YZ}{2}\right)I_r \implies C = Y, \quad D = 1 + \frac{YZ}{2} \tag{52}$$

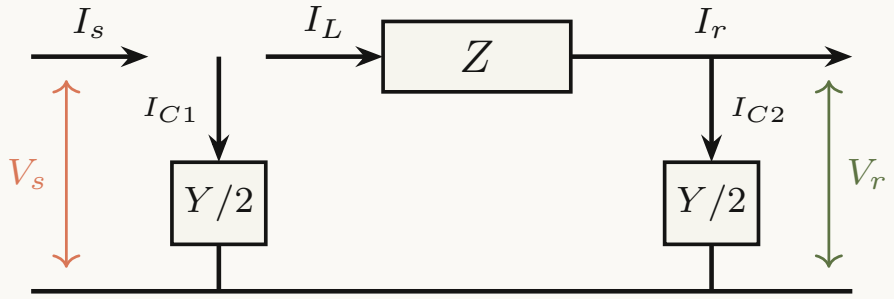
4. Sending-end voltage:

$$\begin{aligned} V_s &= V_C + I_s\left(\frac{Z}{2}\right) = \left[V_r + I_r\left(\frac{Z}{2}\right)\right] + \left[YV_r + \left(1 + \frac{YZ}{2}\right)I_r\right]\frac{Z}{2} \\ &= \left(1 + \frac{YZ}{2}\right)V_r + Z\left(1 + \frac{YZ}{4}\right)I_r \implies A = 1 + \frac{YZ}{2}, \quad B = Z\left(1 + \frac{YZ}{4}\right) \end{aligned} \tag{53}$$

Verification: $AD - BC = \left(1 + \frac{YZ}{2}\right)^2 - Z\left(1 + \frac{YZ}{4}\right)Y = 1 + YZ + \frac{Y^2Z^2}{4} - YZ - \frac{Y^2Z^2}{4} = 1$ ✓

3. Medium Line: Nominal-π Method ($80 - 250$ km)

In the Nominal-π network, series impedance Z is concentrated in the middle branch, and shunt admittance is split equally into two halves ($Y/2$) at sending and receiving terminals.



1. Current in receiving shunt branch: $I_{C2} = \left(\frac{Y}{2}\right)V_r$.
2. Current through series impedance Z: $I_L = I_r + I_{C2} = I_r + \left(\frac{Y}{2}\right)V_r$.
3. Sending-end voltage:

$$V_s = V_r + I_L Z = V_r + \left(I_r + \frac{Y}{2}V_r\right)Z = \left(1 + \frac{YZ}{2}\right)V_r + ZI_r \implies A = 1 + \frac{YZ}{2}, \quad B = Z \tag{54}$$

4. Current in sending shunt branch: $I_{C1} = \left(\frac{Y}{2}\right)V_s = \frac{Y}{2}\left[\left(1 + \frac{YZ}{2}\right)V_r + ZI_r\right]$.
5. Sending-end current:

$$\begin{aligned} I_s &= I_L + I_{C1} = \left(I_r + \frac{Y}{2}V_r\right) + \frac{Y}{2}\left(1 + \frac{YZ}{2}\right)V_r + \frac{YZ}{2}I_r \\ &= Y\left(1 + \frac{YZ}{4}\right)V_r + \left(1 + \frac{YZ}{2}\right)I_r \implies C = Y\left(1 + \frac{YZ}{4}\right), \quad D = 1 + \frac{YZ}{2} \end{aligned} \tag{55}$$

Verification: $AD - BC = \left(1 + \frac{YZ}{2}\right)^2 - Z \cdot Y\left(1 + \frac{YZ}{4}\right) = 1$ ✓

Master Comparison of ABCD Parameters

Model	A	B	C	D
Short Line	1	Z	0	1
Nominal-T	$1+\frac{YZ}{2}$	$Z\left(1+\frac{YZ}{4}\right)$	Y	$1+\frac{YZ}{2}$
Nominal-π	$1+\frac{YZ}{2}$	Z	$Y\left(1+\frac{YZ}{4}\right)$	$1+\frac{YZ}{2}$
Long Line	$\cosh(\gamma l)$	$Z_c \sinh(\gamma l)$	$\frac{1}{Z_c}\sinh(\gamma l)$	$\cosh(\gamma l)$

2.4 Voltage Regulation and Transmission Efficiency

1. Percentage Voltage Regulation ($VR\%$)

Voltage regulation is the percentage change in receiving-end voltage when full rated load at a specified power factor is dropped to zero, keeping sending-end voltage constant.

$$\text{At No Load } (I_r = 0): \quad V_s = AV_{r0} \implies V_{r0} = \frac{V_s}{A} \tag{56}$$

$$\%VR = \frac{|V_{r0}| - |V_r|}{|V_r|} \times 100\% = \frac{\left|\frac{V_s}{A}\right| - |V_r|}{|V_r|} \times 100\% \tag{57}$$

For short lines where $A = 1$, $\%VR = \frac{|V_s|-|V_r|}{|V_r|} \times 100\% \approx \frac{I_r(R\cos\phi_r \pm X\sin\phi_r)}{|V_r|} \times 100\%$ ($+$ for lagging, $-$ for leading).

2. Transmission Efficiency ($\eta\%$)

$$\eta = \frac{\text{Total 3-Phase Active Power Received } (P_r)}{\text{Total 3-Phase Active Power Supplied } (P_s)} \times 100\% \tag{58}$$

$$P_r = \sqrt{3}|V_{r,\text{line}}||I_r|\cos\phi_r = 3|V_{r,\text{ph}}||I_r|\cos\phi_r \tag{59}$$

$$P_s = 3\,\text{Re}\left(V_{s,\text{ph}} \cdot I_s^*\right) = P_r + 3|I_{\text{line}}|^2 R \tag{60}$$

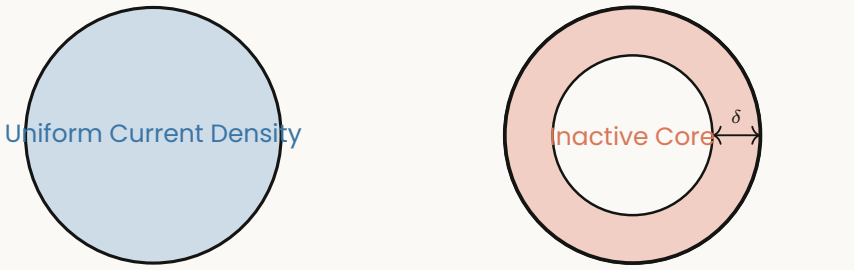
3 Operational Phenomena in Transmission Lines

Syllabus Requirement: Skin Effect & Ferranti Effect – Corona (qualitative study only) – Surge Impedance Loading

3.1 The Skin Effect

Physical Mechanism & Skin Depth

Under direct current (DC), current density J is uniformly distributed over the entire cross-section of a conductor. Under alternating current (AC), the alternating magnetic flux induces concentric eddy currents that oppose current flow at the center and reinforce it at the surface. Consequently, current concentrates in a thin outer layer or extbfskin.



(a) Direct Current (DC) (b) Alternating Current (AC)

The effective depth to which AC current penetrates before decaying to $1/e$ ($\approx 36.8\%$) of its surface value is the extbfskin depth (δ):

$$\delta = \sqrt{\frac{\rho}{\pi f \mu}} = \sqrt{\frac{1}{\pi f \mu \sigma}} \quad \text{[meters]} \tag{61}$$

Factors Influencing Skin Effect

1. **Supply Frequency (f):** Skin effect increases with higher frequency ($\propto \sqrt{f}$).
2. **Conductor Diameter (d):** Increases rapidly with conductor cross-sectional area ($\propto d^2$). Negligible for wire diameters < 1 cm at 50 Hz.
3. **Permeability (μ):** Higher magnetic permeability increases internal flux linkages.
4. **Mitigation:** Using stranded ACSR conductors where the non-conducting steel core occupies the inactive central region, while high-conductivity aluminum occupies the outer perimeter.

3.2 The Ferranti Effect

The Paradox of No-Load Voltage Rise

In 1890, British engineer Sebastian Ziani de Ferranti observed that on medium and long unloaded high-voltage transmission lines, the extbfreceiving-end voltage exceeds the sending-end voltage ($V_r > V_s$).

Mathematical Derivation of Ferranti Rise

Under no-load or light-load conditions, receiving-end load current $I_r \approx 0$. The line draws a pure leading capacitive charging current:

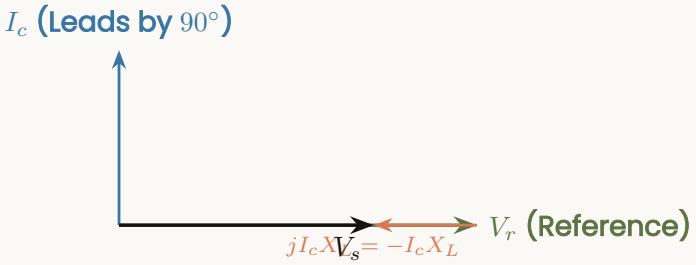
$$I_c = j\omega C V_r \tag{62}$$

This charging current flows through distributed series inductance L. Neglecting line resistance ($R \approx 0$):

$$V_s = V_r + I_c(jX_L) = V_r + (j\omega C V_r)(j\omega L) = V_r - \omega^2 LC V_r = V_r\left(1 - \omega^2 LC\right) \tag{63}$$

Because $\omega^2 LC > 0$, it directly follows that $V_s < V_r$, or:

$$\Delta V = V_r - V_s = V_r \cdot \omega^2 LC \approx \frac{V_r \cdot \omega^2 l^2 L_0 C_0}{2} > 0 \tag{64}$$



Consequences and Mitigation

- **Consequences:** Causes overvoltage stress on insulation, transformer core overexcitation and saturation, and nuisance overvoltage tripping.
- **Mitigation:** Connecting extbfShunt Reactors (inductive coils) at the receiving end substation to absorb excess leading capacitive charging VARs.

3.3 Corona Phenomenon (Qualitative Study)

Definition and Physical Mechanism

Corona is a localized, luminous, self-sustaining dielectric breakdown of air surrounding high-voltage overhead conductors when localized surface electric field gradient exceeds the dielectric breakdown strength of air ($g_0 = 30$ kV/cm peak $= 21.2$ kV/cm rms at NTP).

Characteristics and Manifestations

- Faint pale-violet glow visible in darkness.
- Distinct hissing and crackling sound.
- Production of extbfOzone gas (O_3), recognizable by its pungent odor, which causes chemical corrosion of conductor surfaces.
- Active power loss (I^2R corona loss) and Radio/TV Interference (RI/TVI).

Critical Voltages and Peek's Formula

1. **Critical Disruptive Voltage (V_d):** Minimum phase-to-neutral voltage at which air ionization initiates:
$$V_d = m_0 \cdot g_0 \cdot \delta \cdot r \cdot \ln\left(\frac{d}{r}\right) \quad \text{[kV/phase rms]} \tag{65}$$
where m_0 is conductor surface irregularity factor (1.0 for smooth, $0.8 - 0.87$ for stranded ACSR), $\delta = \frac{3.92b}{273+t}$ is air density correction factor (b in cm of Hg, t in °C).
2. **Visual Critical Voltage (V_v):** Voltage at which the violet glow becomes visually perceptible:
$$V_v = m_v \cdot g_0 \cdot \delta \cdot r \left(1 + \frac{0.3}{\sqrt{\delta r}}\right) \ln\left(\frac{d}{r}\right) \quad \text{[kV/phase rms]} \tag{66}$$
3. **Peek's Corona Power Loss Formula:** Under fair weather conditions, active power loss is:
$$P = \frac{242.2}{\delta}(f + 25)\sqrt{\frac{r}{d}}(V_{\text{ph}} - V_d)^2 \times 10^{-5} \quad \text{[kW/km/phase]} \tag{67}$$

Advantages, Disadvantages, and Corona Mitigation

- **Advantage:** By ionizing surrounding air envelope, corona effectively increases virtual conductor diameter, mitigating steep wavefronts caused by lightning and switching surges (acts as a natural surge relief valve).
- **Disadvantages:** Non-recoverable energy loss, aggressive ozone corrosion, electromagnetic noise disrupting communications.
- **Mitigation:** Increasing conductor diameter using hollow conductors, increasing phase separation d, and deploying extbfbundled conductors.

3.4 Surge Impedance Loading (SIL)

Surge Impedance (Z_c) and Natural Power Loading

The characteristic impedance of a lossless transmission line ($R \approx 0, G \approx 0$) is known as its extbf-Surge Impedance (Z_c):

$$Z_c = \sqrt{\frac{L}{C}} \quad [\Omega] \tag{68}$$

- For typical overhead lines: $Z_c \approx 350 - 450\,\Omega$ (nominal $\approx 400\,\Omega$).
- For underground cables: $Z_c \approx 30 - 50\,\Omega$ (nominal $\approx 40\,\Omega$) due to tightly packed dielectric insulation and high capacitance.

Surge Impedance Loading (SIL): The extbfSurge Impedance Loading (SIL) or *natural power loading* of a transmission line is the electrical power delivered to a purely resistive load equal to its surge impedance ($R_{\text{load}} = Z_c$):

$$\text{SIL} = \frac{V_{\text{L-L}}^2}{Z_c} \quad [\text{MW}] \tag{69}$$

Physical Significance of SIL Operating Regimes

At SIL loading, reactive power generated by line shunt capacitance exactly equals reactive power consumed by series line inductance:

$$Q_C = Q_L \implies \omega C V_{\text{ph}}^2 = \omega L I^2 \implies \frac{V_{\text{ph}}}{I} = \sqrt{\frac{L}{C}} = Z_c \tag{70}$$

1. **Operation at SIL (P = SIL):** Unity power factor along entire line; zero net reactive power exchange; flat voltage profile ($|V_s| = |V_r|$).
2. **Operation Below SIL ($P <$ SIL):** Capacitive generation exceeds inductive absorption ($Q_C > Q_L$); line acts as a net capacitor, causing voltage rise at receiving end (Ferranti rise).
3. **Operation Above SIL ($P >$ SIL):** Inductive consumption exceeds capacitive generation ($Q_L > Q_C$); line acts as a net inductor, causing severe voltage drop.

4 Overhead Line Insulators and String Efficiency

Syllabus Requirement: Insulators – string efficiency – grading (numerical problems)

4.1 Insulator Types and Mechanical Requirements

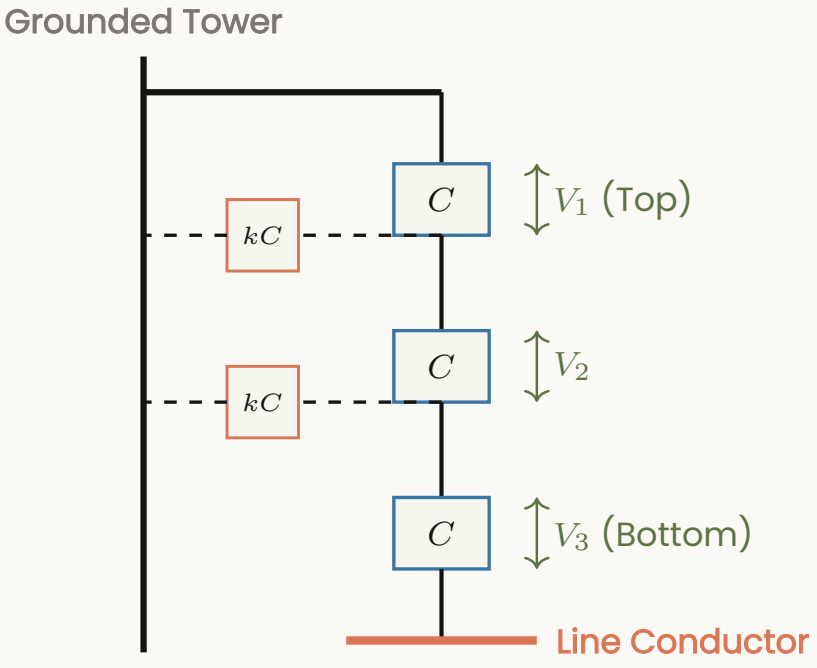
Overhead line conductors are bare, uninsulated cables. Insulators must provide electrical isolation from grounded steel towers while supporting immense mechanical tension from conductor weight, wind pressure, and ice loading.

Insulator Type	Voltage Range	Physical Orientation	Typical Application
Pin Insulator	Up to 33 kV	Rigid vertical mounting	Distribution and sub-transmission lines
Suspension Insulator	>33 kV up to 765 kV	Flexible vertical hanging string of interlocking discs	High-voltage transmission lines; modular (11 kV per standard disc)
Strain / Tension	All HV voltages	Horizontally mounted suspension string	Dead ends, sharp angle towers, river crossings
Shackle / Spool	Low Voltage (<1 kV)	Horizontal or vertical	Low-voltage distribution dead-ends

4.2 Potential Distribution Across a Suspension Insulator String

A suspension string consists of n identical porcelain or toughened glass discs linked by metallic pins.

- Each disc possesses a **mutual capacitance** (C) across its dielectric shell.
- In addition, capacitive coupling exists between each metallic pin and the grounded steel tower body, forming a **shunt capacitance** ($C_1 = kC$), where $k = \frac{C_1}{C} \approx 0.1 - 0.2$.



Mathematical Derivation of Disc Voltages (3-Unit String)

Let V_1, V_2, V_3 be voltages across top, middle, and bottom discs, and ω be angular frequency. Applying Kirchhoff's Current Law (KCL) at each intermediate node:
1. extbfAt Node 1 (between Disc 1 and Disc 2):

$$I_2 = I_1 + i_1 \implies V_2(\omega C) = V_1(\omega C) + V_1(\omega k C) \implies V_2 = V_1(1+k) \tag{71}$$

2. extbfAt Node 2 (between Disc 2 and Disc 3): The potential of Node 2 above ground is $V_1 + V_2$.

$$\begin{aligned} I_3 = I_2 + i_2 &\implies V_3(\omega C) = V_2(\omega C) + (V_1 + V_2)(\omega k C) \\ V_3 = V_2 + k(V_1 + V_2) &= V_1(1+k) + k[V_1 + V_1(1+k)] = V_1\left(1 + 3k + k^2\right) \end{aligned} \tag{72}$$

3. extbfTotal Phase-to-Ground Voltage (V):

$$\begin{aligned} V &= V_1 + V_2 + V_3 = V_1 + V_1(1+k) + V_1(1+3k+k^2) \\ &= V_1\left(3 + 4k + k^2\right) \end{aligned} \tag{73}$$

Key Observation: Non-Uniform Voltage Distribution

Since $k > 0$, it follows that $V_1 < V_2 < V_3$. The disc extbfnearest to the energized line conductor (V_3) experiences the maximum electrical voltage stress and is most prone to dielectric puncture or flashover.

4.3 String Efficiency and Methods of Improvement

String Efficiency (η_{string}): String efficiency is the ratio of total phase voltage across the string to the product of the number of discs and the voltage across the lowest disc:

$$\eta_{\text{string}} = \frac{\text{Total Voltage Across String } (V)}{n \times \text{Voltage Across Disc Nearest to Conductor } (V_n)} \times 100\% \tag{74}$$

Methods of Equalizing Voltage Distribution (Improving η_{string})

1. **Using Longer Cross-Arms:** Increasing tower clearance reduces shunt capacitance C_1, decreasing the ratio $k = C_1/C$. However, tower structural weight and cost limit this approach.
2. **Capacitance Grading of Discs:** Using insulator discs of varying capacitance. Discs with larger mutual capacitance C are placed near conductor end so that $V = \frac{I}{\omega C}$ remains equal across all units.
3. **Static Shield / Guard Ring:** A conductive metallic ring connected directly to the high-voltage conductor surrounding the bottom unit. The ring introduces extbfshielding capacitances (C_2) between the energized conductor and the link pins, injecting currents that neutralize shunt leakage currents to ground, producing **100% uniform voltage distribution**.

5 Comprehensive Numerical Problem Suite

Problem 1: Inductance of a Symmetrically Transposed 3-Φ Single Circuit Line

A 3-phase, 50 Hz, 132 kV transmission line has conductors of diameter 2.0 cm placed at the corners of a triangle with sides 3 m, 4 m, and 5 m. If the line is regularly transposed, calculate: (a) Self GMR, (b) Mutual GMD, (c) Inductance per phase per km, and (d) Inductive reactance per phase for a 100 km line length.

Step-by-Step Solution

1. Conductor radius: $r = \frac{2.0}{2} = 1.0\,\text{cm} = 0.01\,\text{m}$.
2. Self GMR: $D_s = r' = 0.7788 \times 0.01 = \mathbf{0.007788}\,\text{m} = 7.788\,\text{mm}$.
3. Mutual GMD:

$$D_m = \sqrt[3]{D_{12} \cdot D_{23} \cdot D_{31}} = \sqrt[3]{3 \times 4 \times 5} = \sqrt[3]{60} = \mathbf{3.9149}\,\text{m}$$

4. Inductance per phase per meter:

$$L = 2 \times 10^{-7} \ln\left(\frac{D_m}{D_s}\right) = 2 \times 10^{-7} \ln\left(\frac{3.9149}{0.007788}\right)$$
$$= 2 \times 10^{-7} \ln(502.68) = 2 \times 10^{-7} \times 6.22 = \mathbf{1.244 \times 10^{-6}}\,\text{H/m} = \mathbf{1.244}\,\text{mH/km/phase}$$

5. Inductive Reactance for 100 km:

$$X_L = 2\pi f L \times 100 = 2\pi \times 50 \times (1.244 \times 10^{-3}) \times 100 = \mathbf{39.08}\,\Omega\text{/phase}$$

Problem 2: Capacitance of a 3-Φ Transposed Single Circuit Line

For the transmission line in Problem 1, calculate: (a) Line-to-neutral capacitance per km, (b) Line-to-line capacitance per km, and (c) Charging current per km at 132 kV.

Step-by-Step Solution

1. Actual radius: $r = 0.01\,\text{m}$, $D_m = 3.9149\,\text{m}$.
2. Line-to-neutral capacitance:

$$C_n = \frac{2\pi\epsilon_0}{\ln(D_m/r)} = \frac{2\pi \times 8.854 \times 10^{-12}}{\ln(3.9149/0.01)} = \frac{5.5635 \times 10^{-11}}{\ln(391.49)}$$
$$= \frac{5.5635 \times 10^{-11}}{5.97} = \mathbf{9.319 \times 10^{-12}}\,\text{F/m} = \mathbf{0.00932}\,\mu\text{F/km/phase}$$

3. Line-to-line capacitance: $C_{ab} = \frac{C_n}{2} = \mathbf{0.00466}\,\mu\text{F/km}$.
4. Operating phase voltage: $V_{\text{ph}} = \frac{132 \times 10^3}{\sqrt{3}} = 76,210.23\,\text{V}$.
5. Charging current per km:

$$I_c = \omega C_n V_{\text{ph}} = 2\pi \times 50 \times (9.319 \times 10^{-9}\,\text{F/km}) \times 76,210.23 = \mathbf{0.223}\,\text{A/km}$$

Problem 3: Inductance of a Three-Phase Double-Circuit Line

A 3-phase double-circuit line has vertical flat configuration with conductor diameter 2.5 cm. The spacing between vertically adjacent conductors of the same circuit is 4 m. The horizontal distance between the two circuits is 6 m. The phase arrangement is $a-b-c$ on Circuit 1 and $c'-b'-a'$ on Circuit 2. Calculate the inductance per phase per km.

Step-by-Step Solution

1. Coordinates: $a(-3,4), b(-3,0), c(-3,-4)$, and $c'(3,4), b'(3,0), a'(3,-4)$.
2. Conductor parameters: $r = \frac{2.5}{2} = 1.25\,\text{cm} = 0.0125\,\text{m}$, $r' = 0.7788 \times 0.0125 = 0.009735\,\text{m}$.
3. Distances between paired conductors:

$$D_{aa'} = \sqrt{(6)^2 + (8)^2} = \sqrt{36+64} = 10\,\text{m}, \quad D_{bb'} = 6\,\text{m}, \quad D_{cc'} = 10\,\text{m}$$

4. Self GMD of individual phases:

$$D_{sa} = \sqrt{r' \cdot D_{aa'}} = \sqrt{0.009735 \times 10} = 0.3120\,\text{m}$$
$$D_{sb} = \sqrt{r' \cdot D_{bb'}} = \sqrt{0.009735 \times 6} = 0.2417\,\text{m}, \quad D_{sc} = 0.3120\,\text{m}$$
$$D_s = \sqrt[3]{0.3120 \times 0.2417 \times 0.3120} = \mathbf{0.2865}\,\text{m}$$

5. Mutual GMD between phases:

$$D_{ab} = \sqrt[4]{4 \times \sqrt{6^2+4^2} \times \sqrt{6^2+4^2} \times 4} = \sqrt{4 \times 7.2111} = 5.3707\,\text{m}$$
$$D_{bc} = 5.3707\,\text{m}$$
$$D_{ca} = \sqrt[4]{8 \times 6 \times 6 \times 8} = \sqrt{48} = 6.9282\,\text{m}$$
$$D_m = \sqrt[3]{5.3707 \times 5.3707 \times 6.9282} = \mathbf{5.8465}\,\text{m}$$

6. Inductance per phase:

$$L = 2 \times 10^{-7} \ln\left(\frac{D_m}{D_s}\right) = 2 \times 10^{-7} \ln\left(\frac{5.8465}{0.2865}\right)$$
$$= 2 \times 10^{-7} \ln(20.407) = 2 \times 10^{-7} \times 3.0159 = \mathbf{0.603}\,\text{mH/km/phase}$$

Problem 4: ABCD Parameters of a Medium Nominal-T Transmission Line

A 3-phase, 50 Hz, 100 km transmission line has total series impedance $Z = 20 + j40\,\Omega$ and total shunt admittance $Y = j3 \times 10^{-4}\,\text{℧}$. Determine the ABCD constants of the Nominal-T network and verify reciprocity.

Step-by-Step Solution

1. Product YZ:

$$YZ = (j3 \times 10^{-4})(20 + j40) = -0.012 + j0.006$$

2. Parameter A and D:

$$A = D = 1 + \frac{YZ}{2} = 1 + \frac{-0.012 + j0.006}{2}$$
$$= 0.994 + j0.003 = \mathbf{0.9940\angle 0.173^\circ}$$

3. Parameter B:

$$B = Z\left(1 + \frac{YZ}{4}\right) = (20 + j40)\,(0.997 + j0.0015)$$
$$= 19.88 + j39.91\,\Omega = \mathbf{44.59\angle 63.52^\circ}\ \Omega$$

4. Parameter C:

$$C = Y = \mathbf{j3 \times 10^{-4}}\,\mho = \mathbf{0.0003\angle 90^\circ}\,\mho$$

5. Verification of Reciprocity ($AD - BC = 1$):

$$AD - BC = (0.994 + j0.003)^2 - (19.88 + j39.91)(j0.0003)$$
$$= (0.988027 + j0.005964) - (j0.005964 - 0.011973) = \mathbf{1.0000} \quad \checkmark$$

Problem 5: Performance and Voltage Regulation of Nominal-π Line

A 3-phase, 132 kV, 50 Hz line delivers 50 MW at 0.8 power factor lagging at 132 kV. Total line parameters: $Z = 25 + j60\,\Omega$, $Y = j4 \times 10^{-4}\,\mho$. Using the Nominal-$\pi$ model, calculate: (a) Sending-end voltage, (b) Sending-end current, (c) Voltage regulation, and (d) Transmission efficiency.

Step-by-Step Solution

1. Receiving-end phase voltage: $V_r = \frac{132\times10^3}{\sqrt{3}} = 76,210.23\angle 0^\circ$ V.
2. Receiving-end line current:

$$|I_r| = \frac{50 \times 10^6}{\sqrt{3} \times 132 \times 10^3 \times 0.8} = 273.37\,\text{A} \implies I_r = 273.37\angle -36.87^\circ = (218.70 - j164.02)\,\text{A}$$

3. ABCD Parameters (Nominal-π):

$$A = D = 1 + \frac{YZ}{2} = 0.988 + j0.005 = 0.9880\angle 0.29^\circ$$
$$B = Z = 25 + j60\,\Omega = 65.0\angle 67.38^\circ\,\Omega$$
$$C = Y\left(1 + \frac{YZ}{4}\right) \approx j3.976 \times 10^{-4}\,\mho$$

4. Sending-End Voltage (V_s):

$$V_s = AV_r + BI_r = (0.988 + j0.005)(76210.23) + (25 + j60)(218.70 - j164.02)$$
$$= 90604.4 + j9402.6 = \mathbf{91,091.2\angle 5.92^\circ}\ \text{V/phase}$$

$V_{s,\text{line}} = \sqrt{3} \times 91,091.2 = \mathbf{157.77}$ kV.

5. Sending-End Current (I_s):

$$I_s = CV_r + DI_r = (j3.976 \times 10^{-4})(76210.23) + (0.988 + j0.005)(218.70 - j164.02)$$
$$= 216.90 - j130.66 = \mathbf{253.24\angle -31.06^\circ}\ \text{A}$$

6. Percentage Voltage Regulation:

$$\%VR = \frac{|V_s/A| - |V_r|}{|V_r|} \times 100\% = \frac{\frac{91,091.2}{0.988} - 76,210.23}{76,210.23} \times 100\% = \mathbf{20.98\%}$$

7. Transmission Efficiency ($\eta\%$):

$$P_s = 3\,\mathrm{Re}(V_s \cdot I_s^*) = 3 \times |V_s||I_s| \cos(5.92^\circ - (-31.06^\circ))$$
$$= 3 \times 91091.2 \times 253.24 \times \cos(36.98^\circ) = \mathbf{55.29}\,\mathrm{MW}$$
$$\eta = \frac{P_r}{P_s} \times 100\% = \frac{50}{55.29} \times 100\% = \mathbf{90.43\%}$$

Problem 6: String Efficiency of a 3-Disc Suspension Insulator String

In a 3-unit suspension insulator string, the shunt capacitance between each link pin and earth is 15% of the self-capacitance of each unit ($k = 0.15$). If the total working voltage across the string is 33 kV, calculate: (a) Voltage distribution across each disc, and (b) String efficiency.

Step-by-Step Solution

1. Disc Voltage Relationships:

$$V_2 = V_1(1+k) = V_1(1+0.15) = \mathbf{1.15\,V_1}$$
$$V_3 = V_1(1+3k+k^2) = V_1(1+0.45+0.0225) = \mathbf{1.4725\,V_1}$$

2. Total string voltage equation:

$$V = V_1 + V_2 + V_3 = V_1(1+1.15+1.4725) = 3.6225\,V_1 = 33\,\mathrm{kV}$$

3. Voltage across individual discs:

$$V_1 = \frac{33}{3.6225} = \mathbf{9.110}\,\mathrm{kV} \quad \text{(Top Unit)}$$
$$V_2 = 1.15 \times 9.110 = \mathbf{10.476}\,\mathrm{kV} \quad \text{(Middle Unit)}$$
$$V_3 = 1.4725 \times 9.110 = \mathbf{13.414}\,\mathrm{kV} \quad \text{(Bottom Unit nearest to conductor)}$$

4. String Efficiency:

$$\eta_{\text{string}} = \frac{V}{n \times V_3} \times 100\% = \frac{33}{3 \times 13.414} \times 100\% = \frac{33}{40.242} \times 100\% = \mathbf{82.00\%}$$

Problem 7: Capacitance Grading for 100% String Efficiency

A 4-unit suspension insulator string has a shunt-to-mutual capacitance ratio $k = 0.1$ for standard units of mutual capacitance C. If the voltage across every disc is to be made identical ($V_1 = V_2 = V_3 = V_4 = V/4$), calculate the graded mutual capacitances C_1, C_2, C_3, C_4 in terms of C.

Step-by-Step Solution

For equal voltage distribution $V_1 = V_2 = V_3 = V_4 = v$:

1. Top unit: $C_1 = C$.
2. Current in unit 2: $I_2 = I_1 + i_1 \implies v\omega C_2 = v\omega C_1 + v\omega(0.1C) \implies C_2 = C_1 + 0.1C = \mathbf{1.1\,C}$.

3. Current in unit 3: $I_3 = I_2 + i_2 \implies v\omega C_3 = v\omega C_2 + 2v\omega(0.1C) \implies C_3 = C_2 + 0.2C = 1.1C + 0.2C = \mathbf{1.3\,C}$.
4. Current in unit 4: $I_4 = I_3 + i_3 \implies v\omega C_4 = v\omega C_3 + 3v\omega(0.1C) \implies C_4 = C_3 + 0.3C = 1.3C + 0.3C = \mathbf{1.6\,C}$.
5. **Required Grading Ratio:** $C_1 : C_2 : C_3 : C_4 = \mathbf{1.0 : 1.1 : 1.3 : 1.6}$.

Problem 8: Sizing Guard Ring Shielding Capacitance

A 3-unit suspension string has shunt capacitance $C_1 = 0.2\,C$. A guard ring is fitted to equalize the voltages across the bottom two units ($V_2 = V_3$). Find the required shielding capacitance C_2 between the guard ring and the lower intermediate pin.

Step-by-Step Solution

Let voltage across top unit be V_1.

1. At Node 1 (no guard ring connection): $V_2 = V_1(1 + 0.2) = 1.2\,V_1$.
2. At Node 2: The guard ring connected to the line conductor (V) injects current $i_{\text{guard}} = (V - (V_1 + V_2))\omega C_2 = V_3\omega C_2$.
3. KCL at Node 2:

$$I_3 + i_{\text{guard}} = I_2 + i_2 \implies V_3\omega C + V_3\omega C_2 = V_2\omega C + (V_1 + V_2)\omega(0.2C)$$

4. Imposing $V_2 = V_3 = 1.2\,V_1$:

$$1.2V_1C + 1.2V_1C_2 = 1.2V_1C + (V_1 + 1.2V_1)(0.2C)$$

$$1.2C_2 = 2.2 \times 0.2C = 0.44C \implies C_2 = \frac{0.44}{1.2}C = \mathbf{0.367\,C}$$

Problem 9: Surge Impedance Loading of a 400 kV Line

A 400 kV, 3-phase transmission line has series inductance $L = 1.05$ mH/km and shunt capacitance $C = 0.011\ \mu$F/km. Calculate: (a) Surge Impedance (Z_c), (b) Surge Impedance Loading (SIL), and (c) New SIL if bundle conductor modifications decrease L by 10% and increase C by 15%.

Step-by-Step Solution

1. Surge Impedance:

$$Z_c = \sqrt{\frac{L}{C}} = \sqrt{\frac{1.05 \times 10^{-3}}{0.011 \times 10^{-6}}} = \sqrt{95,454.55} = \mathbf{308.96\ \Omega}$$

2. Surge Impedance Loading (SIL):

$$\text{SIL} = \frac{V_{\text{L-L}}^2}{Z_c} = \frac{(400 \times 10^3)^2}{308.96} = \frac{160,000 \times 10^6}{308.96} = \mathbf{517.87\ \text{MW}}$$

3. Modified parameters: $L_{\text{new}} = 0.90 \times 1.05 = 0.945\,\text{mH/km}$, $C_{\text{new}} = 1.15 \times 0.011 = 0.01265\,\mu\text{F/km}$.

$$Z_{c,\text{new}} = \sqrt{\frac{0.945 \times 10^{-3}}{0.01265 \times 10^{-6}}} = \sqrt{74,703.56} = \mathbf{273.32}\,\Omega$$

$$\text{SIL}_{\text{new}} = \frac{(400)^2}{273.32} = \mathbf{585.39}\,\text{MW} \quad (13.04\%\ \text{Capacity Increase})$$

Problem 10: Critical Disruptive Voltage and Visual Corona Calculation

A 3-phase, 220 kV, 50 Hz line has stranded conductors of diameter 2.2 cm spaced equilaterally at 3.5 m. Barometric pressure is 74 cm of Hg and temperature is 30°C. Conductor irregularity factor $m_0 = 0.85$, $m_v = 0.72$, and breakdown strength of air $g_0 = 21.2$ kV/cm (rms). Calculate: (a) Air density correction factor, (b) Critical disruptive voltage, and (c) Visual critical voltage.

Step-by-Step Solution

1. Conductor radius: $r = \frac{2.2}{2} = 1.1$ cm, spacing $d = 350$ cm.
2. Air density correction factor:

$$\delta = \frac{3.92b}{273 + t} = \frac{3.92 \times 74}{273 + 30} = \frac{290.08}{303} = \mathbf{0.9574}$$

3. Critical Disruptive Voltage (Line-to-neutral):

$$V_d = m_0 g_0 \delta r \ln\left(\frac{d}{r}\right) = (0.85) \times (21.2) \times (0.9574) \times (1.1) \times \ln\left(\frac{350}{1.1}\right)$$
$$= 18.976 \times \ln(318.18) = 18.976 \times 5.7626 = \mathbf{109.35}\,\text{kV/phase rms}$$

$V_{d,\text{line}} = \sqrt{3} \times 109.35 = \mathbf{189.40}$ kV.

4. Visual Critical Voltage (Line-to-neutral):

$$V_v = m_v g_0 \delta r \left(1 + \frac{0.3}{\sqrt{\delta r}}\right) \ln\left(\frac{d}{r}\right)$$
$$= 16.074 \times \left(1 + \frac{0.3}{\sqrt{0.9574 \times 1.1}}\right) \times 5.7626 = \mathbf{119.70}\,\text{kV/phase rms}$$

$V_{v,\text{line}} = \sqrt{3} \times 119.70 = \mathbf{207.33}$ kV.

5. **Operational Conclusion:** Operating voltage (220 kV) exceeds both V_d (189.4 kV) and V_v (207.3 kV), meaning extbfvisible corona and power loss will occur.

Problem 11: Corona Power Loss Evaluation using Peek's Formula

For the transmission line in Problem 10, operating at 220 kV line-to-line over a length of 150 km, calculate the total 3-phase active corona power loss under fair weather conditions.

Step-by-Step Solution

1. Operating phase voltage: $V_{\text{ph}} = \frac{220}{\sqrt{3}} = 127.02\,\text{kV}$.
2. Critical disruptive phase voltage: $V_d = 109.35\,\text{kV}$.
3. Voltage excess: $V_{\text{ph}} - V_d = 127.02 - 109.35 = 17.67\,\text{kV}$.
4. Peek's Formula per phase per km:

$$P = \frac{242.2}{\delta}(f+25)\sqrt{\frac{r}{d}}(V_{\text{ph}} - V_d)^2 \times 10^{-5}$$
$$= \frac{242.2}{0.9574}(50+25)\sqrt{\frac{1.1}{350}}(17.67)^2 \times 10^{-5} = \mathbf{3.321}\,\text{kW/km/phase}$$

5. Total 3-Phase Power Loss for $150\,\text{km}$:

$$P_{\text{total}} = 3 \times 3.321 \times 150 = \mathbf{1,494.45}\,\text{kW} \approx \mathbf{1.49}\,\text{MW}$$

Problem 12: Ferranti Rise on an Unloaded Long Transmission Line

A 3-phase, $400\,\text{kV}$, $50\,\text{Hz}$, $300\,\text{km}$ lossless transmission line has distributed parameters $L = 1.0\,\text{mH/km}$ and $C = 0.01\,\mu\text{F/km}$. If the sending-end voltage is held constant at $400\,\text{kV}$, determine: (a) No-load receiving-end voltage, and (b) Rating of shunt reactor required to restore receiving-end voltage to $400\,\text{kV}$.

Step-by-Step Solution

1. Phase propagation constant (β):

$$\beta = \omega\sqrt{LC} = 2\pi \times 50 \times \sqrt{(1.0\times10^{-3})(0.01\times10^{-6})} = 9.934\times10^{-4}\,\text{rad/km}$$
$$\theta = \beta l = (9.934\times10^{-4}) \times 300 = \mathbf{0.2980}\,\text{rad} = \mathbf{17.075}^\circ$$

2. Parameter $A = \cos(\beta l) = \cos(17.075^\circ) = \mathbf{0.9559}$.
3. No-load receiving-end voltage ($I_r = 0$):

$$V_{r0} = \frac{V_s}{A} = \frac{400\,\text{kV}}{0.9559} = \mathbf{418.45}\,\text{kV (Line-to-Line)} \quad (\text{Voltage Rise } = 18.45\,\text{kV})$$

4. Surge Impedance: $Z_c = \sqrt{\frac{L}{C}} = \sqrt{\frac{1.0\times10^{-3}}{0.01\times10^{-6}}} = \mathbf{316.23}\,\Omega$.
5. Shunt Reactor Requirement:

$$X_{\text{reactor}} = \frac{Z_c \sin(\beta l)}{1 - A} = \frac{316.23 \times \sin(17.075^\circ)}{1 - 0.9559} = \frac{316.23 \times 0.2936}{0.0441} = \mathbf{2,105.3}\,\Omega/\text{phase}$$

6. Total 3-Phase Shunt Reactor Rating:

$$Q_{\text{reactor}} = \frac{V_{\text{L-L}}^2}{X_{\text{reactor}}} = \frac{(400\times10^3)^2}{2105.3} = \mathbf{76.0}\,\text{MVAR}$$

6 Quick Revision Sheet & Model Examination Questions

6.1 Master Formula Summary Table for Module 2

Topic / Phenomenon	Governing Mathematical Formula	Physical Significance
Line Inductance	$L = 2 \times 10^{-7} \ln\left(\frac{D_m}{D_s}\right)$ H/m/phase	Self GMR $D_s = 0.7788r$ accounts for internal flux linkages
Line Capacitance	$C_n = \frac{2\pi\epsilon_0}{\ln(D_m/D_{sc})}$ F/m/phase	Uses actual physical radius r (surface charge)
Two-Port Equations	$V_s = AV_r + BI_r, \quad I_s = CV_r + DI_r$	Symmetric ($A = D$), Reciprocal ($AD - BC = 1$)
Nominal-T Model	$A = 1 + \frac{YZ}{2}$, $B = Z(1 + \frac{YZ}{4})$, $C = Y$	Middle lumped shunt admittance
Nominal-π Model	$A = 1 + \frac{YZ}{2}$, $B = Z$, $C = Y(1 + \frac{YZ}{4})$	Shunt admittance split $Y/2$ at ends
Voltage Regulation	$\%VR = \frac{\lvert V_s/A\rvert - \lvert V_r\rvert}{\lvert V_r\rvert} \times 100\%$	Percentage receiving voltage drop on full load
Ferranti Effect	$V_r - V_s \approx \frac{V_r \omega^2 l^2 LC}{2} > 0$	No-load receiving overvoltage; mitigated by shunt reactors
Critical Disruptive	$V_d = m_0 g_0 \delta r \ln(d/r)$ kV/phase	Breakdown of air surrounding high voltage conductors
Surge Impedance	$Z_c = \sqrt{L/C}, \quad \text{SIL} = \frac{V_{LL}^2}{Z_c}$	Flat voltage profile and zero net reactive exchange at SIL
String Efficiency	$\eta = \frac{V}{n \times V_n} \times 100\%$	Voltage non-uniformity across suspension insulator discs

6.2 Model Examination Questions (KTU S5 Pattern)

University Examination Pattern (PCEOT503 Scheme)

The End Semester Examination consists of **Part A** (8 compulsory questions of 3 marks each, 2 from each module) and **Part B** (Module choice questions carrying 9 marks each).

Part A: Short Answer Questions (3 Marks Each)

1. **Q1:** Define *Skin Effect*. Explain why stranded ACSR conductors mitigate skin effect in power transmission.
2. **Q2:** What is *Ferranti Effect*? Draw the phasor diagram under no-load condition and name the equipment used for its mitigation.
3. **Q3:** State the physical significance of *Surge Impedance Loading (SIL)*. What happens when a line operates above SIL?
4. **Q4:** Why is the fictitious radius $r' = 0.7788r$ used in inductance calculation but not in capacitance calculation?
5. **Q5:** Define *String Efficiency* of an insulator string. Why is voltage distribution across suspension discs non-uniform?

6. **Q6:** State the conditions for symmetry and reciprocity in terms of transmission line ABCD parameters.

Part B: Long Answer / Analytical Questions (9 Marks Each)

1. **Q7 (a):** Derive the expression for the loop inductance per phase of a transposed three-phase double-circuit line using the GMD and GMR method. (6 Marks)
   Q7 (b): Explain the advantages of bundled conductors over single solid conductors in EHV lines. (3 Marks)
2. **Q8 (a):** Derive the ABCD parameters for a medium transmission line using the Nominal-π method and show that $AD - BC = 1$. (5 Marks)
   Q8 (b): A 3-phase, 50 Hz, 132 kV transmission line delivers 40 MW at 0.8 power factor lagging. The ABCD parameters are $A = D = 0.98\angle 0.5°$, $B = 60\angle 70° \,\Omega$, $C = 0.0004\angle 90° \,\mho$. Calculate the sending-end voltage and percentage voltage regulation. (4 Marks)
3. **Q9 (a):** Derive the mathematical expressions for the voltage distribution across a 3-unit suspension insulator string and obtain the expression for string efficiency. (5 Marks)
   Q9 (b): Discuss the methods used for improving string efficiency. Explain how a guard ring produces uniform voltage distribution. (4 Marks)

— End of Module 2 Study Notes —